RAHUL BISWAS

Postdoctoral Scholar | Department of Neurology, Weill Institute for Neurosciences
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RESEARCH INTERESTS

Statistical causal inference from neural time series; computational neuroscience; AI and machine learning for biomedical applications; AI-assisted clinical diagnostics and therapy.

EDUCATION

PhD, Electrical and Computer Engineering *June 2024*
University of Washington, Seattle
Thesis: Causal Functional Connectivity from Neural Dynamics

Master of Science, Statistics *March 2023*
University of Washington, Seattle

Master of Statistics *May 2016*
Indian Statistical Institute, Kolkata

Bachelor of Statistics *May 2014*
Indian Statistical Institute, Kolkata

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Postdoctoral Scholar *July 2024 to Present*
Department of Neurology, University of California, San Francisco
Mentor: Dr. Reza Abbasi-Asl

Computational Neuroscience:

- Leading development of the CITS (Causal Inference in Time Series) algorithm for nonparametric causal modeling of high-resolution neural time series.
- Co-leading investigation of chronic pain neural dynamics using unique EEG and ECoG datasets from the UCSF Pain Study, in collaboration with Dr. Prasad Shirvalkar.
- Co-leading analysis of visual cortex functional circuitry and arousal dynamics in mice using calcium imaging and Neuropixels datasets.
- Co-leading AI-based reconstruction of visual experience from fMRI data using biologically constrained generative video models, in collaboration with Dr. Jack Gallant (UC Berkeley).

Computational Clinical Diagnostics:

- Designed the core algorithmic framework for home-based cardiovascular diagnostics (US Provisional Patent, Feb 2026); leading clinical validation at the UCSF Cardiovascular Care and Prevention Center, in collaboration between the Abbasi-Asl Lab and Dr. Julio Lamprea.
- Co-leading development of AI-assisted remote skin lesion diagnostics, in collaboration between the Abbasi-Asl Lab and Dr. Maria Wei, UCSF Dermatology and San Francisco Veterans Affairs.

Research Data Scientist, WOC Federal Appointee *May 2026 – Present*
U.S. Department of Veterans Affairs, Veterans Health Administration
VISN 21-Station 662, San Francisco Office of Research and Development
Collaborating with Dr. Maria Wei, UCSF Dermatology

PhD Researcher
March 2018 to June 2024

Department of Electrical and Computer Engineering, University of Washington, Seattle
Advisor: Dr. Eli Shlizerman

- Developed the Time-Aware PC (TPC) algorithm and statistical network modeling methods for neural time series, published in PLoS Computational Biology and Statistics and Computing.
- Led fMRI studies mapping directional functional connectivity differences in Alzheimer's disease, Mild Cognitive Impairment, and cognitively normal subjects.
- Characterized directed network dynamics in mouse visual cortex using Allen Institute Neuropixels electrophysiology datasets.
- Published 5 peer-reviewed journal articles during doctoral training.

Graduate Research Assistant
July 2015 to May 2016

Statistical and Mathematics Unit, Indian Statistical Institute, Kolkata | Supervisor: Dr. Anil K. Ghosh

Research Intern
May 2015 to July 2015

Department of Statistics, Columbia University, New York City | Supervisor: Dr. Bodhisattva Sen

Biostatistics Research Intern
May 2014 to June 2014

Departments of Epidemiology and Biostatistics, Johns Hopkins Bloomberg School of Public Health | Supervisor: Dr. Xiangrong Kong

Biomedical Signal Processing Intern
May 2013 to December 2013

Systems Science and Informatics Unit, Indian Statistical Institute, Bangalore | Supervisor: Dr. Kaushik Majumdar

HONORS AND AWARDS

- **Best Collaborative Project Prize**, Dynamic Brain Workshop, Allen Institute and University of Washington, Seattle (2018)
- **NSF Travel Award**, Frontier Probability Days Conference, Oregon State University (2018)
- **Departmental Fellowship**, University of Washington, Seattle (2016)
- **INSPIRE Scholarship**, Department of Science and Technology, Government of India (2011 to 2016): Nationally competitive scholarship awarded to the top fraction of science students in India, selected from a national pool.
- **NIUS Fellowship**, Tata Institute of Fundamental Research, India (2012)
- **Certificates of Merit**, Regional Mathematical and Physics Olympiads, India (2010)

PUBLICATIONS

Preprints and Manuscripts Under Review

1. **Biswas, R.,** Senevirathne, H.W., Wang, Y., Zhang, J., Mukherjee, S., & Abbasi-Asl, R. (2026). State-Dependent Organization of Microscale Functional Circuitry in Visual Cortex. bioRxiv preprint. doi:10.64898/2026.03.31.715474
2. **Biswas, R.**, Sripada, S., Mukherjee, S., & Abbasi-Asl, R. (2025). CITS: Nonparametric Statistical Causal Modeling for High-Resolution Neural Time Series. arXiv preprint, doi:10.48550/arXiv.2508.01920

Peer-Reviewed Journal Articles

3. **Biswas, R.**, & Sripada, S. (2025). Alterations in causal functional brain networks in Alzheimer's disease: A resting-state fMRI study. Journal of Dementia and Alzheimer's Disease.
4. Liu, T., Mukherjee, S., & **Biswas, R.** (2024). Tensor Recovery in High-Dimensional Ising Models. Journal of Multivariate Analysis.

5. **Biswas, R.**, & Mukherjee, S. (2024). Consistent Causal Inference from Time Series with PC Algorithm and its Time-Aware Extension. Statistics and Computing.
6. **Biswas, R.**, & Sripada, S. (2023). Causal Functional Connectivity in Alzheimer's Disease Computed from Time Series fMRI Data. Frontiers in Computational Neuroscience.
7. **Biswas, R.**, Thoma, M., & Kong, X. (2023). Functional data analysis to characterize disease patterns in frequent longitudinal data: application to bacterial vaginal microbiota patterns using weekly Nugent scores and identification of pattern-specific risk factors. BMC Medical Research Methodology.
8. **Biswas, R.**, & Shlizerman, E. (2022). Statistical Perspective on Functional and Causal Neural Connectomics: The Time-Aware PC Algorithm. PLoS Computational Biology.
9. **Biswas, R.**, & Shlizerman, E. (2022). Statistical Perspective on Functional and Causal Neural Connectomics: A Comparative Study. Frontiers in Systems Neuroscience.
10. Sarkar, S., **Biswas, R.**, & Ghosh, A.K. (2020). On some graph-based two-sample tests for high dimension, low sample size data. Machine Learning, 109, 279-306.
11. **Biswas, R.**, Khamaru, K., & Majumdar, K. (2014). A peak synchronization measure for multiple signals. IEEE Transactions on Signal Processing, 62(17), 4390-4398.

Selected Conference Proceedings

12. Bhosale, P., & **Biswas, R.** (2022). Relationship between a student's social media use and academic performance. AIP Conference Proceedings, 2471(1), 020008.
13. **Biswas, R.**, & Shlizerman, E. (2020). Neuro-PC: Causal functional connectivity for neural dynamics. BMC Neuroscience, 21(Suppl 1), p.104. 29th Annual Computational Neuroscience Meeting (CNS 2020).
14. **Biswas, R.** (2014). Binary sentiment classification of documents. 7th International Conference of the ERCIM Working Group on Computational and Methodological Statistics, University of Pisa.

PATENTS AND INTELLECTUAL PROPERTY

US Provisional Patent Application *March 2026*

"Multiagent Systems and Methods for Performing Automated Data Science Research with Robustness Assurances"

Inventors: Reza Abbasi-Asl, Rahul Biswas. Assigned to The Regents of the University of California.

US Provisional Patent Application *February 2026*

"Methods and Systems for Assessing Blood Pressure Measurement Technique"

Inventors: Julio A. Lamprea Montealegre, Reza Abbasi-Asl, Rahul Biswas. Assigned to The Regents of the University of California.

PRESENTATIONS AND SEMINARS

Poster Presentations

- Conference on Cognitive Computational Neuroscience (CCN 2026), New York University (accepted, May 2026)
- Computational and Cognitive Neuroscience Meeting (CNCM 2026), University of California, Irvine (submitted, April 2026)
- American Heart Association Scientific Sessions (AHA 2026) (submitted)
- Organization for Human Brain Mapping Annual Meeting, Brisbane, Australia (June 2025)
- Society for Neuroscience Annual Meeting (SfN 2024), Chicago
- 29th Annual Computational Neuroscience Meeting (CNS 2020)

Visiting Seminars

- Brain and Mind Centre, University of Sydney, Australia (June 2025)
- Neuroscience Research Australia (NeuRA) / University of New South Wales, Sydney (June 2025)
- Performance and Expertise Research Centre, Macquarie University, Australia (July 2025)
- Department of Statistics and Data Science, National University of Singapore (2024)
- Chulalongkorn University Neuroscience Center, Bangkok, Thailand (2024)
- Department of Applied Mathematics, Mahidol University International College, Bangkok, Thailand (2024)

   - Indian Statistical Institute, Kolkata (2024)
   - Indian Institute of Science, Bangalore (2024)
   - Department of Psychology, University of Washington, Seattle (January 2023)

Other Oral Presentations

- Dynamic Brain Workshop, Allen Institute and University of Washington, Seattle (August 2018). "Identifying Visual Processing Streams in the Rodent Brain: A Decoding Approach" -- Won Best Collaborative Project Prize
- "Functional Data Analysis in Characterizing Bacterial Vaginosis Patterns and Identifying Pattern-Specific Risk Factors", Johns Hopkins Bloomberg School of Public Health (May 2015)

TEACHING EXPERIENCE

Predoctoral Instructor *June 2022 to August 2022*

Department of Applied Mathematics, University of Washington, Seattle

- Sole instructor for Introduction to Differential Equations and Applications (undergraduate level).
- Supervised a teaching assistant; received student evaluation rating of 4.7 out of 5.0.
- Received a note of appreciation from the Chair of Applied Mathematics.

Predoctoral Teaching Associate *September 2016 to June 2024*

Departments of Statistics and Electrical and Computer Engineering, University of Washington, Seattle

- Assisted in teaching more than 20 courses at the undergraduate, master's, and doctoral levels covering data science, machine learning, and analytical methods.
- Led class sections, managed online discussion forums, and graded coursework.

MENTORING

Current Mentees

- UC Berkeley Undergraduate Research Apprentice Program (URAP), 2025 to present. Mentees: Jonah Cha, Smriti Rangrajan, Sameer Rahman, Joshua Edwards, Jessica Lyu, Aarush Khare.

Former Mentees

- Yuerou Tang, Postbaccalaureate Research Assistant, UC Berkeley (October 2025 to February 2026)
- Hasika Senevirathne, Postdoctoral Researcher, National University of Singapore (2025)
- Yujing Wang, Research Assistant, National University of Singapore (2024 to 2025)
- Tianyu Liu, PhD Research Assistant, National University of Singapore (2023)
- Zhang Jiang, Research Assistant, National University of Singapore (June 2023 to June 2024)
- Yash Chainani, Yunmin Kim, Tishani Weerasuriya, Nicholas Lawrence, Undergraduate Researchers, UC Berkeley Undergraduate Research Apprentice Program (URAP), 2024
- Chavisa Arpavoraruth, Teaching Assistant, University of Washington (June 2022 to August 2022)

SERVICE

Peer Review

- Ad hoc reviewer, journals: Nature Communications, IEEE Transactions on Signal Processing, Frontiers in Neuroscience, Frontiers in Artificial Intelligence, Frontiers in Endocrinology, Frontiers in Global Women's Health, Frontiers in Psychology, Brain Connectivity, Brain and Behavior
- Ad hoc reviewer, conferences and workshops: NeuroAI Workshop at the Association for the Advancement of Artificial Intelligence Annual Conference (AAAI 2026)

Editorial Service

- Editor, Frontiers in Computational Neuroscience

Leadership and Outreach

- President, Science and Consciousness (Registered Staff Organization), University of California, San Francisco (2024 to present)

- President, Science and Consciousness (Registered Student Organization), University of Washington, Seattle (2018 to 2024)

OPEN-SOURCE SOFTWARE

- **CITS algorithm** (Python package): github.com/abbasilab/cits
- **Time-Aware PC algorithm** (Python package, PyPI: timeawarepc): github.com/shlizee/TimeAwarePC. Publicly available since April 2022; independently downloaded and applied by researchers internationally.

TECHNICAL SKILLS

- **Statistical Methods**: Statistical Causal Inference, Causal Discovery, Time Series Analysis, Functional Data Analysis, High-Dimensional Statistics, Machine Learning
- **Programming**: Python, MATLAB, R, SAS, Java, C
- **ML/AI Frameworks**: PyTorch, Transformers, Diffusers, Scikit-learn, PyTorch Lightning
- **Infrastructure**: Git, ClearML, Docker, Kubernetes, Azure, AWS, GCP, HPC schedulers (SGE, Slurm)
- **Data Modalities**: fMRI, Calcium imaging, Neuropixels, EEG, ECoG